§7-2 Trigonometric Integrals

Homework: 3,5,13,16,19,21,29,33,41,67

題型：

(I) $\int \sin^m x \cos^n x \, dx.$

(II) $\int \tan^m x \sec^n x dx$, $\int \cot^m x \csc^n x dx.$

(III) $\int \sin mx \cos nx \, dx$, $\int \sin mx \sin nx \, dx$, $\int \cos mx \cos nx \, dx.$

(I)

i、 m *or* n *is odd*.

- $\begin{cases} u = \sin x & du = \cos x dx \\ u = \cos x & du = (-\sin x) dx \end{cases}$
- $\cos^2 x + \sin^2 x = 1$
- 選擇偶數次方的項為 u ，若 m 和 n 皆為奇數，則 u 任意數。

$$\int \sin^3 x \cos^2 x \, dx = \int \sin^2 x \cos^2 x \sin x \, dx = \int \left(1-u^2\right) u^2 \left(-du\right)$$

$$\int \cos^5 x \sin^4 x \, dx = \int \cos^4 x \sin^4 x \cos x \, dx = \int \left(1-u^2\right)^2 u^4 \, du \, .$$

$$\int \cos^3 x \sin^3 x dx = \begin{cases} \int \cos^3 x \sin^2 x \sin x \, dx = \int u^3 \left(1-u^2\right)\left(-du\right) & \left(u = \cos x\right) \\ \int \cos^2 x \sin^3 x \cos x \, dx = \int \left(1-u^2\right) u^3 du \, . & \left(u = \sin x\right) \end{cases}$$

ii、 m and n are even.

利用$\cos^2 \theta = \dfrac{1+\cos 2\theta}{2}$, $\sin^2 \theta = \dfrac{1-\cos 2\theta}{2}$來降階.

$$\int \sin^4 \theta \, d\theta = \int \left(\frac{1-\cos 2\theta}{2} \right)^2 d\theta = \frac{1}{4} \int \left(1 - 2\cos 2\theta + \cos^2 2\theta\right) d\theta$$

$$= \frac{\theta}{4} - \frac{1}{4}\sin 2\theta + \frac{1}{4}\int \frac{1+\cos 4\theta}{2} d\theta = \frac{3\theta}{8} - \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta + \mathrm{C}.$$

(II)

- $\begin{cases} u = \tan x & du = \sec^2 x \, dx & (a) \\ u = \sec x & du = \sec x \tan x \, dx & (b) \end{cases}$

- $\sec^2 x = \tan^2 x + 1.$

(1)

n *is even* $\Rightarrow$ 利用 (a)

$$\int \tan^3 x \sec^4 x \, dx$$
$$= \int \tan^3 x \sec^2 x \sec^2 x \, dx$$
$$= \int u^3 \left(1+u^2\right) \, du.$$

(2)

m *is odd* $\Rightarrow$ 利用 (b)

$$\int \tan^3 x \sec^3 x \, dx$$
$$= \int \tan^2 x \sec^2 x \tan x \sec x \, dx$$
$$= \int \left(u^2 - 1\right) u^2 \, du.$$

(3)

m is even and n is odd. $\begin{cases} \text{沒有固定方法.} \\ \text{方法之一, 將此型變成} \sin x, \cos x \text{即 } \mathbf{I} \text{ 型.} \end{cases}$

(i) $\displaystyle\int \tan x dx = \int \frac{\sin x}{\cos x} dx = \ln\left|\sec x\right| + \mathrm{C}.$

(ii) $\displaystyle\int \sec x dx = \int (\sec x) \frac{\sec x + \tan x}{\sec x + \tan x} dx = \ln\left|\sec x + \tan x\right| + \mathrm{C}.$

(iii) $\displaystyle\int \sec^{3x} dx$

$$u = \sec x \qquad dv = \sec^2 x dx$$
$$du = \sec x \tan x dx \qquad v = \tan x$$

$$= \sec x \tan x - \int \sec x \tan^2 x \, dx = \sec x \tan x - \int \sec x \left(\sec^2 x - 1\right) dx$$

$$\Rightarrow \int \sec^3 x \, dx = \frac{1}{2}\left[\sec x \tan x + \ln\left|\tan x + \sec x\right|\right] + \mathrm{C}.$$

(III)：將乘法變加法

(1) $\sin A \cos B = \dfrac{1}{2}\left[\sin\left(A-B\right) + \sin\left(A+B\right)\right].$

(2) $\sin A \sin B = \dfrac{1}{2}\left[\cos\left(A-B\right) - \cos\left(A+B\right)\right].$

(3) $\cos A \cos B = \dfrac{1}{2}\left[\cos\left(A-B\right) + \cos\left(A+B\right)\right].$